



QP CODE: 23709200



23709200

Reg No :

Name :

MCA DEGREE EXAMINATIONS, MAY 2023

Third Semester

Master of Computer Application

Core - MCACT301 - MACHINE LEARNING TECHNIQUES

2020 Admission Onwards

FA203E8D

Time: 3 Hours

Maximum: 75 Marks

Part A

*Answer any **ten** questions*

*Each question carries **3** marks*

1. Explain about Machine Learning.
2. What is the meaning of the terms Artificial Intelligence and Machine Learning.
3. In the context of classification problems explain the following terms with examples: (i) hypothesis (ii) hypothesis class
4. Define the terms a) Mean Absolute Error(MAE) b) Mean Squared Error (MSE) c) Root Mean Squared Error(RMSE)
5. Explain the concept of a decision tree with an example.
6. Discuss about binary feature selection.
7. Differentiate between Factor analysis and PCA?
8. Explain about Clustering.
9. What do you mean by Divisive clustering ? Give an example.
10. Differentiate density-reachable and directly density reachable.
11. Compare and contrast between Biological Neurons and Artificial Neurons.
12. What do you mean by Perceptron?

(10×3=30 marks)





Part B

Answer **all** questions

Each question carries **9** marks

13. a) Distinguish between supervised learning and Reinforcement learning. Illustrate with an example.

OR

- b) Define Machine Learning? Explain with example why Machine Learning is important.

14. a) Discuss about learning class in Machine Learning with an example.

OR

- b) Explain about different Univariate Trees with example.

15. a) Explain the method of feature Extraction and feature Selection.

OR

- b) Discuss the concept of principal Component Analysis.

16. a) Explain different types of clustering.

OR

- b) Discuss about the terms i)Voting ii)Bagging iii)Boosting

17. a) Explain about Perceptron neural network and its training algorithm.

OR

- b) Explain about multilayer feedforward neural network with diagram.

(5×9=45 marks)

